

Requirements Document — Telemedicine Platform

Project Overview

The client requires a HIPAA-compliant telemedicine platform enabling patients to consult licensed physicians via secure video calls, chat, and asynchronous messaging. The platform must support appointment scheduling, e-prescriptions, and insurance claim submission.

Target launch: Q3 2026. Expected concurrent users at launch: 5,000. The platform must integrate with Epic EHR and Stripe for payments.

Actors

Patient: books appointments, attends video consultations, views prescriptions.

Doctor: manages availability, conducts consultations, writes e-prescriptions.

Admin: manages user accounts, billing, compliance reports.

Insurance Provider: receives and processes claim submissions via HL7 FHIR API.

Functional Requirements

FR-1 [High] Patients shall register using email or Google SSO and complete identity verification via government-issued ID upload.

FR-2 [High] The system shall support real-time video consultations with end-to-end encryption (WebRTC).

FR-3 [High] Doctors shall be able to issue digital prescriptions compliant with DEA regulations.

FR-4 [Medium] Patients shall receive SMS and email reminders 24 hours and 1 hour before appointments.

FR-5 [Medium] The system shall allow insurance claim submission via HL7 FHIR R4 API.

FR-6 [Low] Patients shall be able to rate and review doctors after consultations.

Non-Functional Requirements

NFR-1 [High] The platform must comply with HIPAA and SOC 2 Type II.

NFR-2 [High] API response time must be under 200ms at the 95th percentile under 5,000 concurrent users.

NFR-3 [Medium] System uptime must be 99.9% monthly.

NFR-4 [Medium] All PHI must be encrypted at rest (AES-256) and in transit (TLS 1.3).

Risks

Risk 1: WebRTC latency on low-bandwidth connections may degrade consultation quality. Mitigation: adaptive bitrate encoding and fallback to audio-only mode.

Risk 2: HIPAA audit failure due to insufficient logging. Mitigation: implement comprehensive audit trails with CloudWatch and quarterly third-party audits.

Risk 3: EHR integration delays from Epic vendor timelines. Mitigation: build abstraction layer supporting both Epic and a local mock during development.